

# A Tutorial on Open-Source Analog-Mixed Signal Chip Design with the ihp-sg13g2 Open-PDK

Simon Dorrer

Harald Pretl

2026-05-11

## Table of contents

|          |                         |          |
|----------|-------------------------|----------|
| <b>1</b> | <b>Overview</b>         | <b>1</b> |
| <b>2</b> | <b>Acknowledgements</b> | <b>2</b> |

### Warning

This documentation is a Work in Progress.

### Important

This repository requires the [IIC-OSIC-TOOLS](#) container with tag 2026.05 or later.

## 1 Overview

This tutorial walks through the complete open-source AMS chip flow for the **ihp-sg13g2** Open-PDK, from RTL and macro generation through padframe integration, physical implementation, verification, and final views.

The example design combines two digital counter macros, two inverter macros, an SRAM macro, analog and digital pad connectivity, and the top-level chip.

## 2 Acknowledgements

This project is funded by the JKU/SAL [IWS Lab](#), a collaboration of [Johannes Kepler University](#) and [Silicon Austria Labs](#).